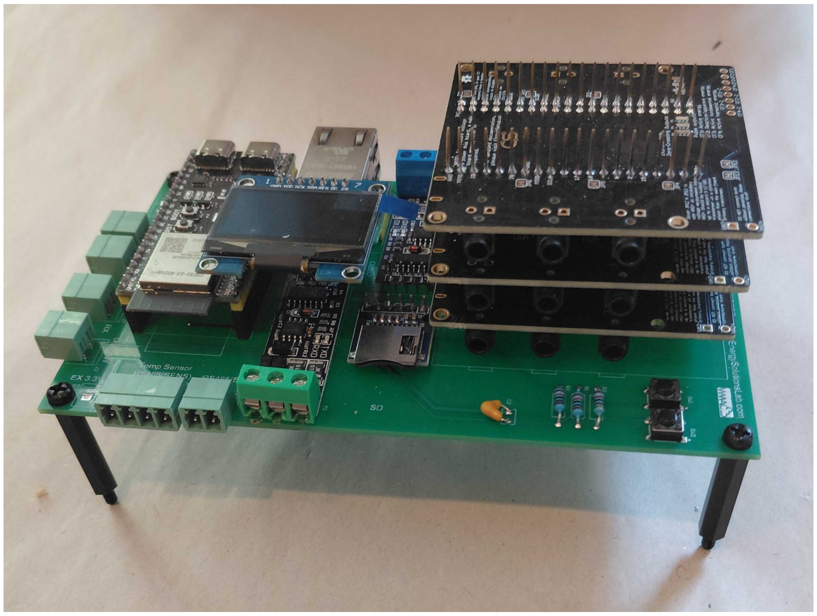
STREET-EMS

Front-of-Meter Feeder Control for the Village Minigrid Street Cabinet



Designing a village minigrid? **STREET-EMS** is the open-source feeder controller for the integration engineer building the **street pedestal cabinet** — coordinating breakers and contactors, metering each branch, and balancing distributed generation across the low-voltage feeder. Front-of-meter, peer-to-peer, and autonomous at the edge.

What Sets STREET-EMS Apart:

- **Cabinet-Ready Control** : drives contactors and breaker coordination from an onboard relay plus an 8-channel I^2C SSR expansion bank for per-branch switching.
- **Open & Vendor-Agnostic** : built on the open-source meshEMS stack; integrates existing meters and DERs over Modbus, with no proprietary lock-in.
- **Edge-Native Balancing** : feeder metering and load balancing run locally on a dual-core edge MCU, with no dependence on cloud or central SCADA.
- **Open-Source Hardware** : released under CERN-OHL-S v2 — schematics, board files and firmware published on GitHub (STREET-EMS).
- **Field-Serviceable** : built from open, off-the-shelf commodity modules, snap-in mounted for fast in-country assembly and repair.

From a single street pedestal, STREET-EMS meters every branch, switches loads via contactors, and coordinates DERs — bringing intelligent, balanced power to the village minigrid feeder.

Intelligent

- Multi-branch metrology: V, I, P, Q, PF, import/export kWh
- Per-channel kW / kWh trip rules for load shedding
- Roadmap: SIP energy-session admission

Interoperable

- 2× RS-485 (Modbus) + CAN 2.0
- 8-ch I^2C SSR for contactors / breakers
- WiFi / BLE & wired Ethernet

Resilient

- Full edge autonomy without backhaul
- Onboard + expansion relay control
- microSD logging | OLED status

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Product Specifications

PRODUCT	
Product name	STREET-EMS (10Power X-EMS, supplied by NESL)
Model / PCB	NESL EMS Controller PCB 886B
Firmware	meshEMS / OpenAMI metering (open source, ∼50k LOC middleware)
License & source	Hardware: CERN-OHL-S v2 | Repository: GitHub, STREET-EMS
Serviceability	Open, off-the-shelf commodity modules | snap-in mounted for fast in-country assembly and repair
COMPUTE	
Processor	Dual-core 32-bit MCU @ 240 MHz | WiFi + Bluetooth LE
Memory	16 MB Flash + 8 MB PSRAM
ASSET COMMUNICATION	
Serial buses	2× RS-485 Modbus: EVC + DTM/sensor ports | 1× CAN 2.0
On-board metering	SPI poly-phase metering, up to 6 CT channels | V, I, P, Q, PF, freq
Local I/O	2× onboard SSR | door-switch input | 2 analog buttons
SUPPORTED MODULES & SENSORS (out of the box)	
Energy meters (Modbus)	Single-phase Modbus energy meters | import/export kWh
Relay expansion	8-channel I^2C solid-state relay (SSR) bank | per-channel kW / kWh trip rules
Environment sensor	Temperature / humidity sensor (Modbus)
DER interoperability	SunSpec models 1 / 11 / 213 | CAN 2.0
NETWORKING	
LAN	10/100 Ethernet | RJ45
Wireless	WiFi 802.11 b/g/n | Bluetooth 5 LE
DISPLAY & STORAGE	
Display & storage	1.3 in OLED display | microSD card slot
POWER & ENCLOSURE	
Operating voltage	5 V DC — USB-C or screw terminals | or 4.5 – 20 V DC | or 8 – 12 V AC (SELV)
Power / dimensions	85 × 55 mm | or Raspberry Pi HAT form factor
STANDARDS & POSITIONING	
Grid position	Front-of-meter (FTM) | LV feeder | street pedestal cabinet
Target standards	IEEE ISV (front-of-meter feeder) | OpenAMI metrology
Capabilities	Peer-to-peer DER & feeder balancing | contactor / breaker switching

Customizable with added hardware modules and open or licensed code stacks. STREET-EMS is an open-source X-EMS product supplied by New Energy Solutions Lab for 10Power.